



RCA & RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

esxi04 Service-Account Credential Mismatch & Jammed-Lock Recovery

Why a "Disconnected" ESXi service account made every Remediate fail and jammed the SDDC Manager lock table — the verified root cause (VCF holds the wrong username for esxi04), and the Broadcom-documented fixes: recreate-to-match (quick) vs decommission/re-commission (clean). Every step KB-cited; the in-place rename is flagged Support-only.

v1.0

VMware Cloud Foundation 9.0.1

RCA & Runbook — esxi04 Service-Account Credential Mismatch + Jammed-Lock Recovery

Prepared by: Virtual Control LLC **Date:** July 29, 2026 **Environment:** VMware Cloud Foundation 9.0.1 (nested homelab), SDDC Manager `sddc-manager.lab.local` **Affected:** ESXi host `esxi04.lab.local` service-account credential; SDDC Manager resource-lock table

Legend: [V] = confirmed by a cited Broadcom KB/TechDoc · [V-LIVE] = confirmed against this environment this session · [UNVERIFIED] = not confirmed by any authoritative source — do not treat as fact.

PART A — Root Cause Analysis

A1. Symptom

- In **SDDC Manager** → **Password Management**, esxi04's ESXi service credential shows **"Disconnected."** [V-LIVE]
- A **Credentials** → **Remediate** on it **FAILS**: *"Validation of password of ESXi host : esxi04.lab.local has failed."* [V-LIVE]
- The failed remediate **jammed 4 resource locks** (`domain/mgmt` , `system/SYSTEM` , `nsx/nsx-vip` , `vcenter/vcenter` , owner empty), blocking further SDDC Manager operations. [V-LIVE]
- The task table also carries a large backlog of stale **IN_PROGRESS** entries (452 observed, dating to May 2026) — the chronic scheduler-jam condition. [V-LIVE]

A2. Evidence gathered (read-only)

Check	Result
Host — <code>esxcli system account list</code> on <code>esxi04</code>	Account present is <code>svc-vcf-esxi04</code> (shell access true) [V-LIVE]
VCF — <code>GET /v1/credentials?resourceType=ESXI</code>	esxi04's record = <code>svc-vcf-esxi01</code> (SSH, SERVICE, id 71633840-...). No <code>svc-vcf-esxi04</code> record exists in VCF. esxi01/02/03 correctly map to svc-vcf-esxi01/02/03 [V-LIVE]
Host connectivity (vCenter)	esxi04 Connected , PoweredOn, healthy (up 41 days, ESXi 9.0.1) [V-LIVE]
<code>esxi04 /var/log/auth.log</code>	No auth failures / lockout entries — the failure is at VCF's validation layer, not the host [V-LIVE]

A3. Ruled out

- **Host down / disconnected** — NO; esxi04 is Connected and serving (the "Disconnected" is a *credential* state, timestamped when the failed remediate ran).
- **Password expiry / account lockout on the host** — NO; no auth failures in the host log, host root SSH works, account exists.
- **Storage / the concurrent memory-pressure event** — SEPARATE incident (see the memory-storm RCA); not the cause of this credential failure.

A4. Root cause (verified)

SDDC Manager holds the WRONG service-account username for esxi04. VCF's credential record for esxi04 is `svc-vcf-esxi01`, but the host's real account is `svc-vcf-esxi04` — and VCF has no `svc-vcf-esxi04` record at all. **Remediate authenticates using the stored username** (`svc-vcf-esxi01`), which **does not exist on esxi04**, so validation fails every time — no number of retries can succeed.

Why the record is wrong: per **Broadcom KB 416351**, this exact symptom occurs when *"the host has been rebuilt/reconfigured and then re-added to the vCenter without going through the decommission/commission process from SDDC Manager."* That matches this environment's **out-of-band vCenter redeploy + host re-add churn**: the host ended up with a correctly-named `svc-vcf-esxi04`, but SDDC Manager's credential record retained the old `svc-vcf-esxi01` label (dated 6/9, never regenerated during the 7/27 inventory resync that fixed esxi01/02/03). [V]

Collateral: the failed Remediate did not release its locks → 4 stale deployment-level locks now block SDDC Manager (the documented lock-jam pitfall). [V-LIVE]

A5. Why the "obvious" fixes cannot work

Action	Why it fails here
Remediate	Password-only; authenticates as the stored <code>svc-vcf-esxi01</code> → ghost account → fails (KB: remediation <i>"addresses password mismatches exclusively, not username discrepancies"</i>) [V]
Rotate	Also uses the stored username; for service accounts it delete/recreates by that name → still wrong target [V]
Update	Documented for user-supplied accounts (e.g., root); service-account passwords can't be set manually. No documented path to rewrite a SERVICE record's username [V] / [UNVERIFIED] for rename
Inventory Sync / Synchronize Inventory Versions	Reconciles versions & inventory objects only — not credential usernames (KB 324050 / 405095) [V]

PART B — Runbook (Remediation)

Order matters. Do NOT start Part B until Phase 0 is complete. Every fix below is a mutating SDDC Manager / host operation and will fail (and can jam more locks) while the lock table is jammed or a live task is running.

Phase 0 — Prerequisites (clear the blockers)

0.1 Let the live bundle download finish (or stop it). A live task violates the lock-clear gate. [V]

0.2 Clear the 4 jammed resource locks — least-destructive first:

1. **KB 432285 — restart SDDC Manager services (PREFERRED, non-destructive):** [V] `ssh vcf@sddc-manager.lab.local → su - /opt/vmware/vcf/operationsmanager/scripts/cli/sddcmanager_restart_services.sh` Wait for full init, then re-check `GET /v1/resource-locks` (expect 0).
2. **KB 339041 / 313431 — targeted release (if restart doesn't clear):** cancel/retry the stuck credentials task (Developer Center → Credentials API, `PATCH / DELETE /v1/credentials/tasks/{taskId}`), then release the lock via API. [V]
3. **Platform-DB cleanup (HEAVY HAMMER, last resort):** the trust-auth `DELETE FROM lock + UPDATE task_meta data SET resolved=true` procedure — see [[RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup-2026-06-09]]. This also clears the **452 stale IN_PROGRESS tasks**. Run only in a known-quiet window, with 0 genuinely-live tasks.

Confirm before proceeding: `GET /v1/resource-locks → 0`; SDDC Manager API responsive; no genuinely-live task.

0.3 Retrieve the stored service-account password (needed for Option A) — KB 327195: [V]

```
ssh vcf@sddc-manager.lab.local → su -  
/opt/vmware/vcf/operationsmanager/scripts/cli/lookup_passwords
```

Find the entry for esxi04's service account (as stored: `svc-vcf-esxi01`). (Confirm exact script path/name against KB 327195 on the live appliance.)

Phase 1 — Choose the fix

Two Broadcom-documented options. They trade *speed against correctness of the name*.

Option A — Recreate-to-match (QUICK; KB 416351 → 314643) [V]

Aligns the **host to VCF's stored record** — i.e. create `svc-vcf-esxi01` on **esxi04** to satisfy the existing record, then Remediate. Clears "Disconnected" and unblocks the patch. **Leaves the cosmetic name wrong** (esxi04's record still says `svc-vcf-esxi01`; no `svc-vcf-esxi04` in VCF).

Steps (SSH to esxi04 as root): (commands as reported by KB 314643 — confirm against the live KB before running)

```
esxcli system account list                # confirm svc-vcf-esxi01 is ABSENT
esxcli system account add --id svc-vcf-esxi01 \
    --password '<lookup_passwords value>' \
    --password-confirmation '<same>'
esxcli system permission set --id svc-vcf-esxi01 --role Admin
esxcli system permission list             # confirm Admin granted
```

Then **SDDC Manager** → **Password Management** → select **esxi04** → **REMEDIATE** with that password → status returns to **ACTIVE**. [V]

Trade-off: esxi04 now has BOTH `svc-vcf-esxi01` (satisfying VCF) and `svc-vcf-esxi04` (the real one from the resync). Functional, Broadcom-documented, but untidy — the record name stays "wrong."

Option B — Decommission + re-commission (CLEAN; KB 416351 intent) [V]

The **only documented way to end up with a correctly-named** `svc-vcf-esxi04` **record**. Hosts should enter/leave a domain **only** via SDDC Manager commission/decommission, which re-mints the correctly-named account + record.

Sequence: 1. **Evacuate esxi04** — vMotion VMs off; **vSAN Full Data Migration**; enter maintenance mode. (*This lab: nsx-manager currently runs on esxi04 — relocate it first, mindful of NSX isolation.*) 2. **Decommission** esxi04 from the workload domain in SDDC Manager (removes its inventory + credential records). 3. **Prepare the host** per the host-prep guide. 4. **Commission** esxi04 back in → creates a correct `svc-vcf-esxi04` service account and credential record.

Trade-off: invasive on a nested single-host vSAN (full data migration + host churn). High effort; do only in a planned window.

Option C — In-place rename of the VCF record — [UNVERIFIED / Support only]

There is **no Broadcom-documented API/UI/DB method** to rename a SERVICE credential's username (`svc-vcf-esxi01` → `svc-vcf-esxi04`) in place. Direct credential-DB edits are **Support-only**. **Do not hand-edit the credentials DB**. If a correct name is required without a decommission, **open a Broadcom case** (reference KB 416351 / 412269).

Phase 2 — Recommendation for THIS lab

Use Option A (recreate-to-match) now; defer Option B. - Option A is Broadcom-documented (KB 314643), **fast, low-risk**, and clears "Disconnected" so the **9.0.2 ESX patch is unblocked** — which is the functional goal. - The wrong *name* is **cosmetic** — it does not affect function; the account works. - Option B (decommission/re-commission) is the *clean* fix but is **heavy** on a nested single-host vSAN (full data migration, NSX relocation).

Reserve it for a planned rebuild window if you want the name truly correct. - **Never** attempt the in-place DB rename (Option C).

Phase 3 — Verify

- **Host:** `esxcli system account list` on esxi04 shows the account VCF expects, with **Admin** (`esxcli system permission list`). [V]
- **VCF:** `GET /v1/credentials?resourceType=ESXi` → esxi04 credential no longer **Disconnected**; Password Management shows **Active**. [V-LIVE check]
- **Locks:** `GET /v1/resource-locks` → **0**.
- **Precheck:** re-run the 9.0.2 precheck → the esxi04 "**ESXi Host checks**" / service-account item clears.

Part C — Sources (Broadcom, verified 2026-07-29)

- **KB 416351** — ESXi `svc-vcf-esxi_####` disconnected in Password Management (root cause: re-add without decommission/commission; fix → KB 314643) — knowledge.broadcom.com/external/article/416351/
- **KB 314643** — Re-create missing SDDC Manager service account on an ESXi host (the `esxcli system account add + permission` steps) — [.../314643/](https://kb.broadcom.com/314643/)
- **KB 327195** — Retrieve service-account credentials (`lookup_passwords`) — [.../327195/](https://kb.broadcom.com/327195/)
- **KB 412269** — Troubleshooting account disconnects in Password Management — [.../412269/](https://kb.broadcom.com/412269/)
- **KB 401681** — Remediate fails "validation of password of ESXi host has failed" (custom password-policy variant) — [.../401681/](https://kb.broadcom.com/401681/)
- **KB 396739** — `svc-vcf` disconnected via SSH host-key mismatch (→ KB 316028 `fixHostkeys.py`) — [.../396739/](https://kb.broadcom.com/396739/)
- **KB 432285** — Clear stale locks via `sddcmanager_restart_services.sh` (preferred) — [.../432285/](https://kb.broadcom.com/432285/)
- **KB 339041 / 313431** — Password-manager related-resource locks / targeted release — [.../339041/](https://kb.broadcom.com/339041/), [.../313431/](https://kb.broadcom.com/313431/)
- **KB 324050 / 405095 / 419613** — inventory sync scope (versions/objects, not credentials; brownfield host restrictions) — [.../324050/](https://kb.broadcom.com/324050/), [.../405095/](https://kb.broadcom.com/405095/), [.../419613/](https://kb.broadcom.com/419613/)
- **TechDocs** — Remediate Passwords · Updating SDDC Manager Passwords · Look up account credentials (VCF 9.0 Fleet Management)
- **Developer** — SDDC Manager Credentials API (developer.broadcom.com/xapis/sddc-manager-api/latest/credentials/)

Verification caveat (per research): exact command strings from KB 314643 / 432285 / 327195 are as reported by the KB pages via web fetch — **confirm against the live article before running**. The single most important [UNVERIFIED] point: **no Broadcom-documented in-place method exists to rename a SERVICE credential's username** — that correction is achieved only by decommission/re-commission (Option B) or via Broadcom Support (Option C).

Document Information

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Revision History

Version	Date	Notes
1.0	2026-07-29	Initial RCA + runbook. Verified root cause (VCF stores <code>svc-vcf-esxi01</code> for esxi04; host has <code>svc-vcf-esxi04</code> ; no <code>svc-vcf-esxi04</code> in VCF — KB 416351 out-of-band re-add pattern). Documented fixes: Option A recreate-to-match (KB 314643, quick), Option B decommission/re-commission (clean), Option C in-place rename (UNVERIFIED/Support). Added Phase 0 lock recovery (KB 432285 service restart → 339041/313431 → DB-cleanup runbook). All KB-sourced; rename flagged Support-only.

Document End